Task 4 Presentation

Ferreira (2017) Investigation I

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Equations 1 and 2

We begin with a couple of definitions. $Y_t(s)$ are our observed values. $M_t(s)$ are covariates. β are the usual parameters, and ϵ is our spatio-temporal dependencies.
ψ are the MA(∞) representations, and $\eta_{t-j}(s)$ are a 'sequence of temporally independent and spatially Gaussian processes, with $E(\eta_s(s)) = 0, \forall s \in D$', and covariance structure to follow.

$$Y_t(s) = M_t(s)\beta + \epsilon_t(s)$$

$$\epsilon_t(s) = \sum_{k=0}^{\infty} \psi_j \eta_{t-j}(s)$$

Covariance (1/4)

For this slide we take a look at the covariance structure using the assumptions of the previous slide.

$$
\begin{aligned}
\operatorname{cov}(Y_t(s), Y_{t'}(s')) &= \operatorname{cov}(M_t(s)\beta + \epsilon_t(s), M_{t'}(s')\beta + \epsilon_{t'}(s')), \quad \text{by (1)} \\
&= \operatorname{cov}(\epsilon_t(s), \epsilon_{t'}(s')), \quad \text{as } M_t(s), \beta \text{ are deterministic} \\
&= \operatorname{cov}(\sum_{j=0}^{\infty} \psi_j \eta_{t-j}(s), \sum_{k=0}^{\infty} \psi_k \eta_{t'-k}(s') \quad \text{by (2)} \\
&= \sum_{j=0}^{\infty} \sum_{k=0}^{\infty} \psi_k \psi_j \operatorname{cov}(\eta_{t-j}(s), \eta_{t'-k}(s'))
\end{aligned}
$$

by definition 2.5 in the class notes

Covariance Continued (2/4)

Now we have the basic breakdown we are more familiar with and can apply the assumptions we set in slide 1. Starting with our Temporal Independence, we have that $\text{cov}(\eta_t(s), \eta_{t'}(s)) = 0$ when $t \neq t'$. We then solve for $t - j = t + h - k \rightarrow k = j + h$ for a single summation as usual and we get:

$$\text{cov}(\epsilon_t(s), \epsilon_{t'}(s') = \sum_{j=0}^{\infty} \psi_j \psi_{j+h} \, \text{cov}(\eta_{t-j}(s), \eta_{t-j}(s'))$$

Covariance Continued (3/4)

Now we consider the element $\text{cov}(\eta_{t-j}(s), \eta_{t-j}(s'))$. Since time is fixed, we have only the spatial dependence. As $\eta_t(s)$ is defined as spatially stationary we have that:

$$\text{cov}(\eta_{t-j}(s), \eta_{t-j}(s')) = C^{\eta}(||s - s'||; \theta) \tag{3}$$

We note that $C^{\eta} : [0, \infty) \to \mathcal{R}$ such that $C^{\eta} \circ || \cdot || : \mathcal{R}^2 \to \mathcal{R}$. Ferreira define the use of euclidean distance, so it is inherently isotropic. We will return to the exact formula of the spatial covariance term later.

Covariance Continued (4/4)

Now we return to the general formula for $Y_t(s)$.

$$\begin{aligned}
\operatorname{cov}(Y_t(s), Y_{t'}(s')) &= \sum_{i,j=0}^{\infty} \psi_j \psi_i \operatorname{cov}(\eta_{t-j}(s), \eta_{t-j}(s')) \\
&= C^{\eta}(\xi) \sum_{j=0} \psi_j \psi_{j+|t-t'|} \\
&= C^{\eta}(\xi)\kappa(h)
\end{aligned}$$

This has the neat property of being the product of a purely spatial and a purely temporal component. The method of determining $C^{\eta}(\xi)$ and $\kappa(h)$ will depend highly upon the spatial and temporal model used.

Determining parameters

Now that we have identified the basic covariance structure, we will look at specific temporal instances, and then we will observe the Matern covariance model for spatial components.

- ARMA(p,q) (1)
- ARFIMA(p,d,q) (2)
- Matern Covariance (3)

ARMA(p,q) covariance

For this slide, we examine some common ARMA(p,q) covariance structures. (Question, the class notes include a σ_w^2 in the covariance of AR(1) but the paper does not, curious. Example 7.3)

$$\kappa(h) = \mathrm{Cov}(\mathrm{X_t}, \mathrm{X_{t+h}}) = \frac{\sigma_w^2 \phi^h}{1-\phi^2} \qquad \text{AR(1)(pg 77)}$$

$$\kappa(h) = \begin{cases} (1+\theta^2)\sigma_w^2 & h = 0 \\ \theta\sigma_w^2 & h = 1 \\ 0 & h > 1 \end{cases} \qquad \text{MA(1)(pg 82)}$$

$$\kappa(h) = \begin{cases} \sigma_w^2 \frac{1+2\theta\phi+\theta^2}{1-\phi^2} & h = 0 \\ \sigma_w^2 \frac{(1+\theta\phi)(\phi+\theta)}{1-\phi^2} & h \geq 1 \end{cases} \qquad \text{ARMA(1,1)(pg 96)}$$

Footnote link:
Mikosch et. al. (1995) Didn't seem particularly relevent

ARFIMA(p,d,q) covariance

Next up, example 2 from the paper which covers ARFIMA models

$$\kappa(h) = \sigma^2 \frac{\Gamma(1-2d)\Gamma(h+d)}{\Gamma(1-d)\Gamma(d)\Gamma(h+1+d)} \qquad h > 0, d \in (0, 1/2)$$

Kokoszka and Taqqu (1995)

Matern Spatial Covariance

The Matern Spatial Covariance function is apparently something that is well known. Note that K_v is the Bessel function of the second order of v. Whatever that means

$$C^{\eta}(\xi;\theta) = \frac{\sigma^2}{2^{v-1}\Gamma(v)}(2\sqrt{v}\rho\xi)^v K_v(2\sqrt{v}\rho\xi) \tag{4}$$

$$\xi \geq 0, \boldsymbol{\theta} = (\sigma^2, \rho, v)^T, \rho > 0, v \geq 0, \sigma^2 > 0 \tag{1}$$

Matern (1986) Textbook, not publically available
Porcu et. al. (2023?)

Prior State Space Models

Updating our equation to match the text book from last time, let $y_{1:t}$ be a set of observed data points. Let $s_{1:t}$ be the set of **unobservable** states.

Predict Step

$$\hat{s}_{t|t-1} = \phi \hat{s}_{t-1|t-1} + \gamma u_t \qquad \text{Predicted state}$$

$$P_{t|t-1} = \phi P_{t-1|t-1} \phi^T + Q \qquad \text{Prediction Cov.}$$

Update Step

$$\hat{s}_{t|t} = \hat{s}_{t|t-1} + K_t(y_t - A_t \hat{s}_{t|t-1} - \Gamma u_t) \qquad \text{Updated state}$$

$$P_{t|t} = [I - K_t A_t] P_{t|t-1} \qquad \text{Update pred. cov.}$$

$$K_t = P_{t|t-1} A_t^T [A_t P_{t|t-1} A_t^T + R]^{-1} \qquad \text{Kalman Gain}$$

State Space Model Introduction

We now break down the state space model in the paper and relate it to what we know.

$$Y_t(s) = \begin{bmatrix} G_t(s) & M_t(s) \end{bmatrix} \begin{bmatrix} X_t(s) \\ \beta_t(s) \end{bmatrix} + W_t(s)$$

- $Y_t(s)$ is the observation matrix
- $G_t(s)$ is an observation operator
- $M_t(s)$ is a vector of predetermined variables
- $\begin{bmatrix} X_t(s) \\ \beta_t(s) \end{bmatrix}$ is the state vector
- $W_t(s)$ is the noise with variance R_w

State Prediction

Now onto the kalman filter, previously we had the state as $\hat{s}_{t|t-1} = \phi\hat{s}_{t-1|t-1} + \gamma u_t$, but it is now represented as:

$$\begin{bmatrix} X_{t+1}(s) \\ \beta_{t+1}(s) \end{bmatrix} = \begin{bmatrix} F_t(s) & \mathbf{0} \\ \mathbf{0} & I_p \end{bmatrix} \begin{bmatrix} X_t(s) \\ \beta_t(s) \end{bmatrix} + \begin{bmatrix} H \\ \mathbf{0} \end{bmatrix} V_t \tag{5}$$

Note that V_t is spatially colored and temporally white and Gaussian with mean 0 and covariance function $Cov(V_t(s), V_t(s') = C^V(\xi; \boldsymbol{\theta})$ and that $V_t(s)$ is uncorrelated with $W_t(s)$

After Action

- Vanessa's dad wrote the koko 1995 paper
- Long Range Dependence and Self-Similarity by Taqqu and Pipiras contains half question's answers
- Spatial statistics textbook by Michael Stein could be good too.

Thank you!